Diploma in Electronics / Electronics & Communication Engineering • Semester 3

Fundamentals of C Programming

Introduces C programming structure, data types, input/output, operators, control statements, arrays, pointers, strings and functions for solving real-world problems.

Course Code: 3045 | **Category: Program Core** | **Credits: No Credit** | **L:T:P = 4:0:0** | **Programming Theory**

COURSE OVERVIEW

Fundamentals of C Programming (3045)

This is a standalone digital textbook, revision guide, workbook and answer guide prepared from the uploaded official syllabus PDF.

Student-friendly summary

Introduces C programming structure, data types, input/output, operators, control statements, arrays, pointers, strings and functions for solving real-world problems.

Study tip: ഓരോ module-ഉം separate ആയി പഠിക്കുക. Definitions, diagrams, formulas/procedures, and expected answers English technical terms ഉപയോഗിച്ച് practise ചെയ്യുക.

Course Metadata

ITEM	DETAILS
Course Code	3045
Base Course Code	3045
Course Title	Fundamentals of C Programming
Programme / Department	Diploma in Electronics / Electronics & Communication Engineering
Semester	3
Course Category	Program Core
Credits	No Credit
L:T:P	4:0:0
Periods / semester	60
Subject Type	Programming Theory

Course Objectives

- Provide basic knowledge in C programming.
- Use control blocks, looping, string manipulation and functions to solve real world problems.
- Utilize C programming for real-life programming tasks.

Prerequisite Knowledge

- Basic principles and theorems of Engineering Mathematics - Mathematics I & II
- Computer, operating system, internet applications and basic Python programming - Introduction to IT Systems Lab

Course Outcomes

CO	DESCRIPTION	HOURS	LEVEL
CO 1	Make use of programming concepts with C programming.	13	Applying
CO 2	Make use of iterative control structures and arrays in programs.	15	Applying
CO 3	Develop programs using pointers and strings.	15	Applying
CO 4	Make use of functions to solve problems effectively.	15	Applying

Module Map

1. **Module 1:** C programming basics (13 h)
2. **Module 2:** Control structures and arrays (15 h)
3. **Module 3:** Pointers and strings (15 h)
4. **Module 4:** Functions (15 h)

How to study this subject

- Understand syntax using small examples.
- Trace every program line by line.
- Practise algorithms before coding.
- Run at least two test cases for each program.
- Revise common compiler errors.

Exam preparation method

- Write algorithm first, then program.
- Use correct indentation and comments.
- Mention sample input/output.
- Check loops, arrays, pointers and functions carefully.

Quick Navigation Cards

Key Points Bank - Fundamentals of C Programming

Quick reference cards for revision. Use this only after reading the module explanation.

Program structure

```
Header -> main() -> declarations -> statements -> return
```

Meaning: Standard C program layout.

Where used: All programming answers.

Common mistake: Writing statements outside functions.

Input/output

```
scanf("%d", &x); printf("%d", x);
```

Meaning: Read and display values.

Where used: Basic programs.

Common mistake: Forgetting & in scanf for normal variables.

if-else

```
if(condition){...} else {...}
```

Meaning: Decision making.

Where used: Flow-control problems.

Common mistake: Using = instead of ==.

for loop

```
for(init; condition; update){...}
```

Meaning: Count-controlled repetition.

Where used: Series/array problems.

Common mistake: Wrong loop boundary.

Array

```
int a[10];
```

Meaning: Stores same-type values in continuous memory.

Where used: List/matrix problems.

Common mistake: Accessing a[10] in 10-size array.

Pointer

```
int *p = &x;
```

Meaning: Stores address of a variable.

Where used: Pointers and call by reference.

Common mistake: Dereferencing uninitialised pointer.

String functions

```
strlen, strcpy, strcat, strcmp
```

Meaning: Built-in string operations.

Where used: String handling.

Common mistake: Not including string.h.

Function

```
return_type name(parameters){...}
```

Meaning: Reusable block of code.

Where used: Modular programs.

Common mistake: Mismatch between prototype and definition.

MODULE 1 • 13 HOURS

C programming basics

Use C program structure, keywords, variables, constants, data types, I/O, operators and type casting.

Learning outcome

Use C program structure, keywords, variables, constants, data types, I/O, operators and type casting.

Malayalam support: ആദ്യം meaning മനസ്സിലാക്കുക, പിന്നെ English technical terms ഉപയോഗിച്ച് answer എഴുതുക.

Important terms

main() keyword variable constant data type scanf printf operator type casting

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M1.01	Understand structure of a C program	1	Understanding
M1.02	Use keywords, variables, constants, data types and type qualifiers	3	Applying
M1.03	Experiment with basic input and output functions	3	Applying
M1.04	Use arithmetic, relational, conditional, logical, bit-wise and assignment operators	4	Applying
M1.05	Apply type casting on data types	2	Applying

Topic-wise explanation

Structure of a C program: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Keywords, variables and constants: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Data types and type qualifiers: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Output and input functions: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Arithmetic operators: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

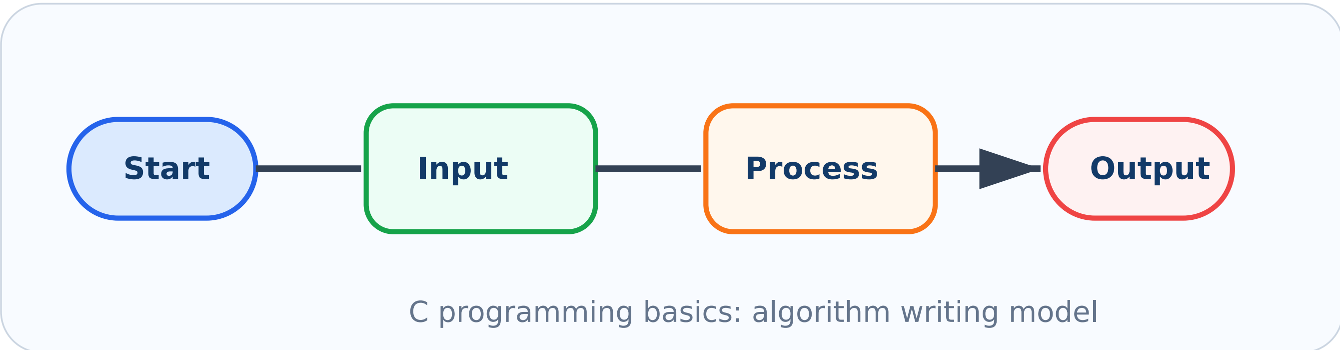
Relational and logical operators: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Increment/decrement and conditional operators: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Assignment and bitwise operators: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Type casting: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Visual block / diagram idea



Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

```
#include <stdio.h>
int main(void){
    int a,b;
    scanf("%d%d", &a, &b);
    printf("Sum = %d", a+b);
    return 0;
}
```

Explain header file, main function, variable declaration, input, processing and output.

Common mistakes

- Missing semicolon or braces.
- Using wrong format specifier in scanf/printf.
- Array index out of range.
- Using uninitialised pointer or variable.
- Not showing sample input and output.

Expected Questions

M1-Q1 Explain the main concept of C programming basics with syntax or format.

M1-Q2 Write a C program / code fragment related to Structure of a C program.

M1-Q3 Compare two important concepts from C programming basics.

M1-Q4 Find and correct common errors in a C program using C programming basics.

M1-Q5 Write the algorithm and output for a problem from C programming basics.

M1-Q6 List applications and exam tips for C programming basics.

Module checklist

✓ Syntax and keywords understood.

✓ Program can be traced manually.

✓ At least two sample inputs tested.

✓ Common errors checked.

✓ Output is written exactly.

MODULE 2 • 15 HOURS

Control structures and arrays

Use decision statements, loops, unconditional control statements, one-dimensional arrays, two-dimensional arrays and matrix operations.

Learning outcome

Use decision statements, loops, unconditional control statements, one-dimensional arrays, two-dimensional arrays and matrix operations.

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Important terms

if-else · switch · while · do-while · for · break · continue · array · matrix

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M2.01	Experiment with control structures	2	Applying
M2.02	Make use of looping structures	2	Applying
M2.03	Develop programs with unconditional control statements	2	Applying
M2.04	Experiment with one-dimensional array operations	3	Applying
M2.05	Make use of two-dimensional array declaration	3	Applying
M2.06	Make use of matrix operations	3	Applying

Topic-wise explanation

if, if-else and nested if: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

switch statement: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

while, do-while and for loops: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Nested loops: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

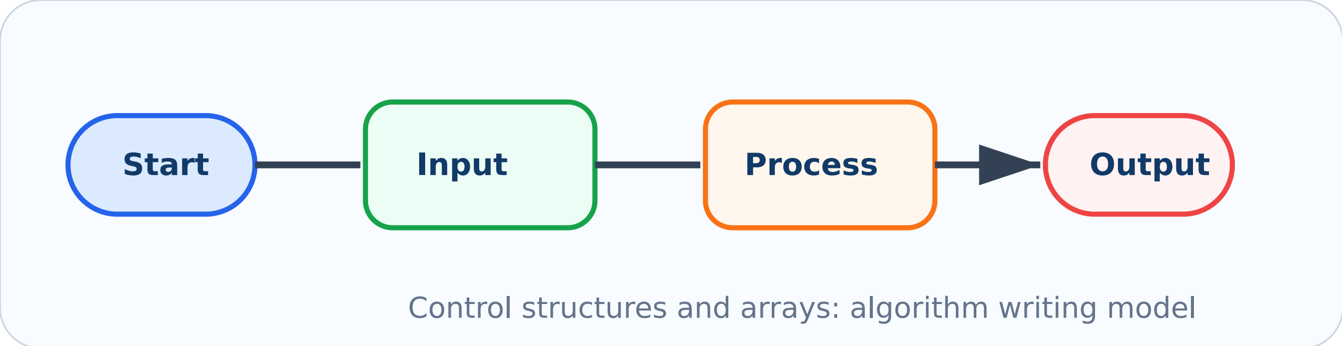
break, continue and goto: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

1D array insertion, deletion, searching and sorting: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

2D array declaration: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Matrix addition and multiplication: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Visual block / diagram idea



Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

```
#include <stdio.h>
int main(void){
    int a,b;
    scanf("%d%d", &a, &b);
    printf("Sum = %d", a+b);
    return 0;
}
```

Explain header file, main function, variable declaration, input, processing and output.

Common mistakes

- Missing semicolon or braces.
- Using wrong format specifier in scanf/printf.
- Array index out of range.
- Using uninitialised pointer or variable.
- Not showing sample input and output.

Expected Questions

M2-Q1 Explain the main concept of Control structures and arrays with syntax or format.

M2-Q2 Write a C program / code fragment related to if, if-else and nested if.

M2-Q3 Compare two important concepts from Control structures and arrays.

M2-Q4 Find and correct common errors in a C program using Control structures and arrays.

M2-Q5 Write the algorithm and output for a problem from Control structures and arrays.

M2-Q6 List applications and exam tips for Control structures and arrays.

Module checklist

✓ Syntax and keywords understood.

✓ Program can be traced manually.

✓ At least two sample inputs tested.

✓ Common errors checked.

✓ Output is written exactly.

MODULE 3 • 15 HOURS

Pointers and strings

Use pointers, pointer arithmetic, strings using pointers, gets/puts and built-in string handling functions.

Learning outcome

Use pointers, pointer arithmetic, strings using pointers, gets/puts and built-in string handling functions.

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Important terms

pointer | address | dereference | pointer arithmetic | string | gets | puts | strlen

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M3.01	Develop concepts of pointers in programming	2	Applying
M3.02	Use pointer arithmetic operations	3	Applying
M3.03	Develop programs for accessing strings using pointers	3	Applying
M3.04	Experiment with gets() and puts() functions	3	Applying
M3.05	Use built-in string handling functions	4	Applying

Topic-wise explanation

Basics of pointers: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Pointer arithmetic addition, subtraction, increment and decrement: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

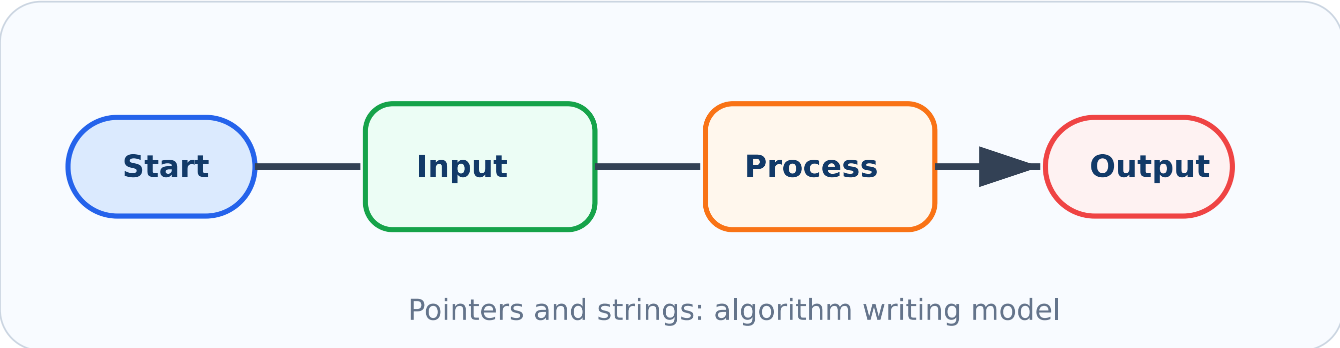
Pointer comparison: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Accessing strings using pointers: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

gets() and puts() functions: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

String handling functions: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Visual block / diagram idea



Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

```
#include <stdio.h>
int main(void){
    int a,b;
    scanf("%d%d", &a, &b);
    printf("Sum = %d", a+b);
    return 0;
}
```

Explain header file, main function, variable declaration, input, processing and output.

Common mistakes

- Missing semicolon or braces.
- Using wrong format specifier in scanf/printf.

- Array index out of range.
- Using uninitialised pointer or variable.
- Not showing sample input and output.

Expected Questions

M3-Q1 Explain the main concept of Pointers and strings with syntax or format.

M3-Q2 Write a C program / code fragment related to Basics of pointers.

M3-Q3 Compare two important concepts from Pointers and strings.

M3-Q4 Find and correct common errors in a C program using Pointers and strings.

M3-Q5 Write the algorithm and output for a problem from Pointers and strings.

M3-Q6 List applications and exam tips for Pointers and strings.

Module checklist

✓ Syntax and keywords understood.

✓ Program can be traced manually.

✓ At least two sample inputs tested.

✓ Common errors checked.

✓ Output is written exactly.

MODULE 4 • 15 HOURS

Functions

Define functions, write user-defined function structure, use function declarations/calls/arguments/return type and recursion.

Learning outcome

Define functions, write user-defined function structure, use function declarations/calls/arguments/return type and recursion.

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Important terms

function

prototype

argument

return type

call by value

call by reference

recursion

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M4.01	Outline functions, types and advantages	2	Understanding
M4.02	Illustrate structure of user-defined functions	3	Understanding
M4.03	Use function declaration, call, arguments and return type	3	Applying
M4.04	Construct different types of functions	2	Applying
M4.05	Build programs with call by value and call by reference	3	Applying
M4.06	Use recursive functions	2	Applying

Topic-wise explanation

Definition of functions: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Structure of user-defined functions: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Function prototype and function call: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Arguments and return type: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

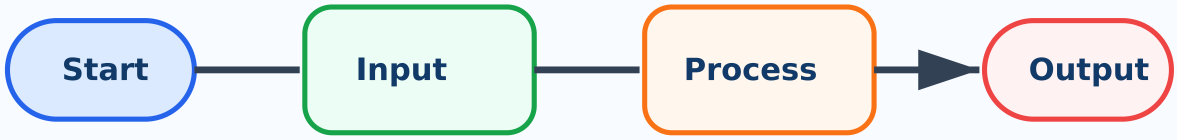
Types of functions: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Call by value: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Call by reference: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Recursive functions: Learn the syntax first, then write small programs and test with sample inputs. Keep variable names meaningful and explain the program flow before writing code.

Visual block / diagram idea



Functions: algorithm writing model

Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

```
#include <stdio.h>
int main(void){
    int a,b;
    scanf("%d%d", &a, &b);
    printf("Sum = %d", a+b);
    return 0;
}
```

Explain header file, main function, variable declaration, input, processing and output.

Common mistakes

- Missing semicolon or braces.
- Using wrong format specifier in scanf/printf.
- Array index out of range.
- Using uninitialised pointer or variable.
- Not showing sample input and output.

Expected Questions

M4-Q1 Explain the main concept of Functions with syntax or format.

M4-Q2 Write a C program / code fragment related to Definition of functions.

M4-Q3 Compare two important concepts from Functions.

M4-Q4 Find and correct common errors in a C program using Functions.

M4-Q5 Write the algorithm and output for a problem from Functions.

M4-Q6 List applications and exam tips for Functions.

Module checklist

✓ Syntax and keywords understood.

✓ Program can be traced manually.

✓ At least two sample inputs tested.

✓ Common errors checked.

✓ Output is written exactly.

QUICK REVISION

One-page Revision Summary

Use this page in the last 24 hours before exam or lab evaluation.

Module-wise must-know points

1. **Module 1 - C programming basics:** Revise program skeleton, data types, I/O and operator precedence.
2. **Module 2 - Control structures and arrays:** Practise flow control syntax, loop tracing, array operations and matrix programs.
3. **Module 3 - Pointers and strings:** Know address/dereference, pointer arithmetic rules and string functions with examples.
4. **Module 4 - Functions:** Practise function syntax, argument passing and recursive program tracing.

Important definitions / keywords

keyword variable constant data type type qualifier operator type casting control statement loop array pointer string function call by value call by reference recursion

Important formulas / key points

- Header -> main() -> declarations -> statements -> return
- scanf("%d", &x); printf("%d", x);
- if(condition){...} else {...}
- for(init; condition; update){...}
- int a[10];
- int *p = &x;
- strlen, strcpy, strcat, strcmp
- return_type name(parameters){...}

Common exam mistakes

- Missing semicolon or braces.
- Using wrong format specifier in scanf/printf.
- Array index out of range.
- Using uninitialised pointer or variable.
- Not showing sample input and output.

Last-minute checklist

✓ Write algorithm first, then program.

✓ Use correct indentation and comments.

✓ Mention sample input/output.

✓ Check loops, arrays, pointers and functions carefully.

Diagram / table list

- C program structure diagram
- Decision flowchart

- Loop flowchart
- 1D/2D array memory representation
- Pointer/address diagram
- Function call flow
- Recursion stack concept

MASTER ANSWER KEY

Master Answer Key - matched with Expected Questions

Every answer below matches the module expected questions exactly. Attempt first, then verify here.

M1-Q1 - Explain the main concept of C programming basics with syntax or format.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for C programming basics, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M1-Q2 - Write a C program / code fragment related to Structure of a C program.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for C programming basics, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M1-Q3 - Compare two important concepts from C programming basics.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for C programming basics, and display output clearly. For exam answers, write **algorithm -> program -> sample**

input/output -> explanation. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M1-Q4 - Find and correct common errors in a C program using C programming basics.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for C programming basics, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M1-Q5 - Write the algorithm and output for a problem from C programming basics.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for C programming basics, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M1-Q6 - List applications and exam tips for C programming basics.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for C programming basics, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M2-Q1 - Explain the main concept of Control structures and arrays with syntax or format.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Control structures and arrays, and display output clearly. For exam answers, write **algorithm -> program -> sample**

input/output -> explanation. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M2-Q2 - Write a C program / code fragment related to if, if-else and nested if.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Control structures and arrays, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M2-Q3 - Compare two important concepts from Control structures and arrays.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Control structures and arrays, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M2-Q4 - Find and correct common errors in a C program using Control structures and arrays.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Control structures and arrays, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M2-Q5 - Write the algorithm and output for a problem from Control structures and arrays.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Control structures and arrays, and display output clearly. For exam answers, write **algorithm -> program -> sample**

input/output -> explanation. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M2-Q6 - List applications and exam tips for Control structures and arrays.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Control structures and arrays, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M3-Q1 - Explain the main concept of Pointers and strings with syntax or format.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Pointers and strings, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M3-Q2 - Write a C program / code fragment related to Basics of pointers.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Pointers and strings, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M3-Q3 - Compare two important concepts from Pointers and strings.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Pointers and strings, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M3-Q4 - Find and correct common errors in a C program using Pointers and strings.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Pointers and strings, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M3-Q5 - Write the algorithm and output for a problem from Pointers and strings.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Pointers and strings, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M3-Q6 - List applications and exam tips for Pointers and strings.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Pointers and strings, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M4-Q1 - Explain the main concept of Functions with syntax or format.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Functions, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M4-Q2 - Write a C program / code fragment related to Definition of functions.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Functions, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M4-Q3 - Compare two important concepts from Functions.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Functions, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M4-Q4 - Find and correct common errors in a C program using Functions.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Functions, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M4-Q5 - Write the algorithm and output for a problem from Functions.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Functions, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output -> explanation**. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

M4-Q6 - List applications and exam tips for Functions.

Start with the required header files and `main()`. Declare variables with correct data types, read inputs using suitable input statements, process the logic for Functions, and display output clearly. For exam answers, write **algorithm -> program -> sample input/output ->**

explanation. Mention common errors such as missing semicolon, wrong format specifier, uninitialised variable, array index out of range and pointer misuse.

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